

AWS Serverless File Upload Notification System

Step-by-Step Implementation Guide (Without Screenshots)

Project Objective

The objective of this project is to build a serverless application that automatically sends an email notification whenever a file is uploaded to an Amazon S3 bucket. The workflow uses Amazon S3, AWS Lambda, Amazon SNS, IAM, and CloudWatch.

Architecture

File Upload → Amazon S3 → S3 Event Notification → AWS Lambda → Amazon SNS → Email Notification

Step 1: Create an S3 Bucket

Open the AWS Management Console and navigate to Amazon S3. Click 'Create bucket', enter a globally unique bucket name, select your preferred AWS Region, keep Block Public Access enabled, and create the bucket.

Step 2: Create an SNS Topic

Navigate to Amazon SNS, create a Standard topic, provide a topic name such as 'S3UploadNotification', and create the topic.

Step 3: Create an Email Subscription

Open the SNS topic, create a subscription using the Email protocol, enter your email address, and confirm the subscription using the confirmation email sent by AWS.

Step 4: Create an IAM Role

Open IAM, create a new role for Lambda, and attach the AWSLambdaBasicExecutionRole and AmazonSNSFullAccess managed policies. Save the role.

Step 5: Create a Lambda Function

Create a new Lambda function using Python runtime. Attach the IAM role created earlier. Replace the default code with Python code that reads S3 event details and publishes a message to the SNS topic.

Step 6: Configure the Lambda Trigger

Add an S3 trigger to the Lambda function. Select your S3 bucket and configure the trigger for Object Created events.

Step 7: Deploy the Lambda Function

Deploy the function after verifying the code and the SNS Topic ARN.

Step 8: Test the Project

Upload any file to the S3 bucket. Amazon S3 generates an Object Created event, which automatically invokes the Lambda function.

Step 9: Verify Email Notification

Check your email inbox. Amazon SNS should send an email containing details of the uploaded file, including bucket name and object key.

Monitoring

Open Amazon CloudWatch Logs to verify that the Lambda function executed successfully and troubleshoot any errors if necessary.

Conclusion

This project demonstrates a complete serverless, event-driven architecture using AWS managed services. It automates file upload notifications without requiring server management and provides a scalable and cost-effective cloud solution.

aws

Search

[Alt+S]

Asia Pacific (Mumbai)

beingamaan (865122443570)

beingamaan

Amazon S3

Buckets

General purpose buckets

Directory buckets

Table buckets

Vector buckets

Files

File systems

Access management and security

Access Points

Access Points for FSx

Access Grants

IAM Access Analyzer

Storage management and insights

Storage Lens

Batch Operations

Storage

Amazon S3

Store and retrieve any amount of data from anywhere

Amazon S3 is an object storage service that offers industry-leading scalability, data availability, security, and performance.

Create a bucket

Every object in S3 is stored in a bucket. To upload files and folders to S3, you'll need to create a bucket where the objects will be stored.

Create bucket

How it works

Introduction to Amazon S3 | Amazon Web Services

Pricing

With S3, there are no minimum fees. You only pay for what you use. Prices are based on the location of your S3 bucket.

Estimate your monthly bill using the [AWS Simple Monthly Calculator](#)

[View pricing details](#)

CloudShell

Feedback

Create Mobile App

© 2025 Amazon Web Services, Inc. or its affiliates

Privacy

Terms

Cookie preferences

aws

Search

[Alt+S]

Asia Pacific (Mumbai)

beingamaan (865122443570)

beingamaan

Amazon S3

Buckets

amaan-s3-bucket123

amaan-s3-bucket123

Info

Objects

Metadata

Properties

Permissions

Metrics

Management

File systems - new

Access Points

Objects (0)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

< 1 >

Name

Type

Last modified

Size

Storage class

No objects

You don't have any objects in this bucket.

Upload

CloudShell

Feedback

Console Mobile App

© 2026, Amazon Web Services, Inc. or its affiliates.

Privacy

Terms

Cookie preferences

aws

Search

[Alt+S]

Asia Pacific (Mumbai)

beingamaan (865122443570)

beingamaan

Amazon SNS > Dashboard

Amazon SNS

<

Dashboard

Topics

Subscriptions

▼ Mobile

Push notifications

Text messaging (SMS)

📘 New Feature

Amazon SNS now supports High Throughput FIFO topics. [Learn more](#)

Dashboard

Resources for ap-south-1

Topics

3

Platform applications

0

Subscriptions

3

▼ Overview of Amazon SNS

Application-to-application (A2A)

Amazon SNS is a managed messaging service that lets you decouple publishers from subscribers. This is useful for application-to-application messaging for microservices, distributed systems, and serverless applications. [Learn more](#)

Dead-letter queue

If an endpoint is unavailable, messages can be held in a dead-letter queue for analysis or reprocessing

AWS Lambda

Amazon SQS

Amazon Kinesis Data Firehose

HTTP/HTTPS

Email

CloudShell Feedback Console Mobile App

© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

aws

Search

[Alt+S]

Asia Pacific (Mumbai)

beingamaan (865122443570)

beingamaan

Amazon SNS

Topics

amaan-sns

Amazon SNS

Dashboard

Topics

Subscriptions

Mobile

Push notifications

Text messaging (SMS)

New Feature

Amazon SNS now supports High Throughput FIFO topics. [Learn more](#)

amaan-sns

Edit

Delete

Publish message

Details

Name

amaan-sns

ARN

arn:aws:sns:ap-south-1:865122443570:amaan-sns

Display name

-

Type

Standard

Topic owner

865122443570

Subscriptions

Access policy

Delivery policy (HTTP/S)

Delivery status logging

Encryption

Tags

Integrations

Subscriptions (1)

Edit

Delete

Request confirmation

Confirm subscription

Create subscription

Search

< 1 >

ID

Endpoint

Status

Protocol

1202a23d-0cb6-48ce-85f9-f37af6...

amaanalikhankino@gmail.com

Confirmed

EMAIL

https://ap-south-1.console.aws.amazon.com/console/home?region=ap-south-1

© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences



Simple Notification Service

Subscription confirmed!

You have successfully subscribed.

Your subscription's id is:

arn:aws:sns:ap-south-1:865122443570:amaan-sns:1202a23d-0cb6-48ce-85f9-f37af6e886e7

If it was not your intention to subscribe, [click here to unsubscribe](#).

aws

Search

[Alt+S]

beingamaan (865122443570)

beingamaan

Lambda > Functions > amaan-lambda

amaan-lambda

Throttle

Copy ARN

Actions

Function overview

Diagram

Template

amaan-lambda

Layers (0)

S3

Add trigger

Export to Infrastructure Composer

Download

Function ARN

arn:aws:lambda:ap-south-1:865122443570:function:amaan-lambda

Description

-

Last modified

1 hour ago

Code

Test

Monitor

Configuration

Aliases

Versions

Code source

Open in Visual Studio Code

Update

Info

Tutorials

Choose a tutorial

Learn how to implement common use cases in AWS Lambda.

Create a simple web app

In this tutorial you will learn how to:

- Build a simple web app, consisting of a Lambda function with a function URL that outputs a webpage
- Invoke your function through its function URL

Learn more

Start tutorial

CloudShell

Feedback

Console Mobile App

© 2026, Amazon Web Services, Inc. or its affiliates.

Privacy

Terms

Cookie preferences

Upload succeeded

For more information, see the [Files and folders](#) table.

After you navigate away from this page, the following information is no longer available.

Summary

Destination

s3://amaan-s3-bucket123

Succeeded

1 file, 11.0 MB (100.00%)

Failed

0 files, 0 B (0%)

Files and folders (1 total, 11.0 MB)

Find by name

< 1 >

Name	Folder	Type	Size	Status	Error
Java-Unit-1-One-Shot-Notes (1)...	-	application/pdf	11.0 MB	Succeeded	-

